

Aaron Stone — 290 Revolution Drive (Somerville)

Routine

Generated from the [exercise](#) corpus (adversarially-verified claims · muscle/length physiology · set-structure RCTs · the Renaissance-Periodization MEV→MAV→MRV landmark system) against the **290 Revolution Drive (Somerville)** apartment-gym equipment menu and Aaron's constraint profile. **Not medical advice** — the shoulder work here is therapeutic strengthening *within available ROM*, not rehab prescription. Keep any physio/ortho follow-up and honor the red-flag kill switch below.

Who / where / what

Field	Value
Lifter	Aaron Stone
Gym	290 Revolution Drive (Somerville) — apartment gym (DBs cap 75 lb ; Olympic bar ~570 lb ceiling; Smith ~550 lb; cables/FTS; selectorized machines; no leg-press / hack-squat / dedicated calf or adduction machine)
Split	4-Day Upper/Lower — Upper A / Lower A / Upper B / Lower B; every muscle trained ~2×/week
Goal	Dual strength + physique — one genuine 5–8 rep strength anchor per session, the rest 8–20 for hypertrophy
Headline constraint	Bilateral shoulder instability (prior dislocation; rotator-cuff/labral involvement; LIMITED external-rotation ROM, <i>no pain</i>). The single thing to avoid is the apprehension position — abduction combined with end-range external rotation.

Bilateral-shoulder programming rules (always on): (1) **Neutral-grip / front-of-body pressing only** — no behind-the-neck press or pulldown, no wide pronated end-range work. (2) **Ban the apprehension position** — never load abduction + end-range external rotation. (3) **No deep dips, no forced end-range stretch under load** — full but *pain-free* ROM, hard stop at the first sign of instability. (4) **Controlled, non-ballistic tempo** (~2 s eccentrics, no push-press/jerk). (5) **Intensity techniques (drop sets / myo-reps) stay OFF every shoulder-loaded lift** — confined to machine/cable isolation. (6) **A programmed rotator-cuff + scapular-stability block runs TWICE weekly** (~6 sets/upper day = 12/week) at 1–2 RIR, never to grinding failure. (7) **Posterior-girdle volume ≥ pressing volume** (rear delt 16, back 18 vs. anterior delt 10) as a structural-balance safeguard. (8) **The protection IS the training** — upper-body volume is set *generous-but-recoverable*, not cut to "protect" the joint; sedentariness worsens his symptoms, in-ROM training resolves them.

A separate document covers Aaron's other gym (Planet Fitness). This file is the 290 Revolution Drive (Somerville) routine only.

Philosophy & schedule

FREQUENCY / SPLIT. A 4-day Upper/Lower hits every muscle ~2×/week — the established hypertrophy frequency floor (Schoenfeld 2016: ≥2× beats 1×) and a sufficient strength dose for intermediates — while keeping each session under the ~6–8 quality-set/muscle per-session ceiling (Weightology). That ceiling is exactly *why* Aaron's high weekly upper volume is split across two upper days rather than crammed into one.

LOADING (the load/rep continuum, honestly). Hypertrophy is load-independent from ~30–85% 1RM at matched effort (Lopez/Schoenfeld — the "heavy load required for hypertrophy" and "failure required" claims are **refuted**), so most work sits **8–15 reps at moderate load and moderate effort**. A genuine **5–8 rep strength anchor** per session (one neutral-grip press on each upper day; one knee-dominant and one hip-dominant compound on the lower days) captures the real 1–6 rep / 80–100% 1RM strength-zone specificity. Net mix lands ~60–70% of sets in a personal 6–15 sweet spot, ~15–20% heavier (6–8), ~15–20% lighter (15–20 metabolite work) — the SBS sweet-spot spread.

EFFORT / RIR (autoregulated by exercise class and block week). Compound strength anchors **3–4 RIR early-block → 2–3 RIR late** (Helms/Robinson — failure on heavy compounds is slightly inferior for strength and ~triples acute neuromuscular fatigue vs 3 RIR; reserve 0–1 RIR for an end-of-block top set only, and only on the patterns Aaron presses pain-free). Primary hypertrophy compounds **2–3 RIR, last set ~1**. Isolation **1–2 RIR, final set to 0–1** (the closer-to-failure hypertrophy premium is cheap on low-systemic-stress lifts; mandatory 0–1 RIR on any *light-load* high-rep isolation set because motor-unit recruitment is recruitment-late at low loads). RIR ramps **down ~4 → ~1 across each 4–5 week accumulation block** (Martikainen — equal gains at lower average session-RPE), then a deload at 3–6 RIR.

PROGRESSION. Double-progression within each exercise's rep band: keep load fixed and add reps across sessions until you hit the top of the band on all sets at target RIR, then add the smallest load increment and drop back to the bottom. **DBs cap at 75 lb** — once a DB lift maxes out, progress via reps / added sets / tempo / structure, never grip-width creep on the shoulder. Within each block, ramp volume from ~MEV (8–12 sets/muscle) into the MAV zone (12–20) by adding ~1–3 sets/week to lagging/priority muscles (lateral delt, rear delt, back, triceps), then deload to MV (~6 sets) when performance stalls and restart at MEV.

SET-STRUCTURE LOGIC (no technique out-grows volume-equated straight sets). Straight sets anchor every shoulder-stakes compound (lowest fatigue cost, best load/technique control) — ~70% of total weekly sets. Time-efficient structures are deployed only where they buy efficiency or fatigue-management without risking the joint: **antagonist supersets** (Burke 2025 RCT — identical hypertrophy/strength at ~36% less session time, and they conveniently pair scapular pull against pressing); **myo-reps** and **capped machine drop sets** on machine/cable isolation only; **cluster sets** for velocity-maintained heavy pressing; **reverse-pyramid** as an optional autoregulatory swap on the *lower-body* anchors only. Pre-exhaust agonist supersets and any high-fatigue technique on overhead/dip/behind-the-body movements are **avoided**.

CONSTRAINT INTEGRATION (always on). Every press is neutral-grip and front-of-body; the apprehension position (behind-neck press/pulldown, wide end-range overhead) is banned; no deep dips or forced end-range stretch; tempo is controlled and non-ballistic; intensity techniques stay off shoulder-loaded lifts; posterior-girdle volume ≥ pressing volume; the rotator-cuff + scapular block runs twice weekly at full dose as a programmed priority. **Red-flag kill switch** halts any set on a subluxation/dislocation event, sharp pain, a slipping/instability sensation, arm numbness/tingling, or sudden strength loss — dull working-muscle pump/fatigue is expected and fine.

Day	Session	Targets	Anchor
Upper A — Horizontal-Emphasis (push-bias) + shoulder-health block #1	Upper	chest, lats/mid-back, lateral + rear delt, biceps, triceps, rotator-cuff/scap #1	Neutral-grip DB flat press, 4 × 6–8 @ 2–4 RIR (heavier strength anchor)
Lower A — Knee-Dominant (quad-emphasis)	Lower	quads, glutes, hamstrings (long-length), adductors, gastroc (standing) , core	High-bar back squat, 4 × 5–8 @ 2–4 RIR (knee-dominant anchor)

Day	Session	Targets	Anchor
Upper B — Vertical-Emphasis (front-of-body) + shoulder-health block #2	Upper	anterior + lateral + rear delt, upper chest, lats (front), biceps, triceps, rotator-cuff/scap #2	Neutral-grip seated DB shoulder press, 4 × 5–8 @ 2–4 RIR (overhead anchor)
Lower B — Hip-Dominant (hamstring/glute-emphasis)	Lower	hamstrings (both heads), glutes (long + short length), quads (2°), adductors, soleus (seated) , core	Romanian deadlift, 3 × 5–8 @ 2–3 RIR (hip-dominant anchor)

Run it as a 4-on / rotate schedule (e.g. Mon Upper A · Tue Lower A · Thu Upper B · Fri Lower B, or any 4 days that give each upper/lower pair ~48 h between repeats). Order within each upper day runs cuff/scap activation first (primes scapular stability, warms the joint), then the big press+row, then delts/arms/scap finisher.

Weekly volume (generous-but-recoverable; shoulder targets dosed therapeutically)

Muscle	Weekly sets	Note
Chest (pectoralis major)	14 (Upper A ~6 heavy neutral press + row-incidentals, Upper B ~3 incline + indirect)	2×/wk. Anchors are neutral-grip flat (Upper A) and incline neutral DB (Upper B) — front-of-body, controlled tempo, stretch kept full but pain-free (no forced wide-grip deep stretch).
Back / lats (lats, teres major, rhomboids, mid/lower trap)	18 (Upper A ~9 horizontal, Upper B ~6–9 vertical)	2×/wk. Chest-supported + seated cable rows (horizontal, Upper A); close neutral-grip pulldown to chest (vertical, Upper B). Neutral grips substituted for any wide pronated end-range pull. Kept ≥ pressing volume (RULE 7).
Anterior deltoid	10 (Upper B ~5 direct overhead + Upper A pressing indirect)	Mostly the dedicated neutral overhead-press day (Upper B) plus indirect from horizontal pressing. Lowest of the delt heads on purpose — pressing is conservative, laterals/rears drive delt growth.
Lateral deltoid	18 (Upper A ~5, Upper B ~9, split)	2×/wk, high therapeutic priority — dosed generously, not cut . Cable lateral raise to ~shoulder height only (out of the abduction+ER apprehension zone); constant-tension loaded stretch at the bottom.
Posterior (rear) deltoid	16 (Upper A ~5, Upper B ~8 + row-incidentals)	2×/wk. Face pulls + cross-body rear lateral raise. Kept ≥ pressing volume as the structural-balance safeguard for the anterior capsule (RULE 7).
Rotator cuff + scapular stability (infraspinatus/teres minor/subscap + serratus/lower trap)	12 (~6/upper day)	Programmed priority, twice weekly, never an afterthought . Cable external rotation (elbow at side), serratus protraction, face pulls, overhead scap shrug. 1–2 RIR, controlled, never to grinding failure. This volume does NOT start low or get cut on a deload — it is the literal mechanism that resolves Aaron's symptoms (RULE 8).
Biceps + brachialis	14 (Upper A ~5, Upper B ~3 + pull-incidentals)	2×/wk. EZ-bar curl (Upper A) + lengthened incline DB curl (Upper B). Fully shoulder-neutral (elbows beside the torso).
Triceps	14 direct (Upper A ~5, Upper B ~3 + heavy pressing indirect)	2×/wk. Rope pushdown + overhead long-head stretch. Triceps specifically benefit from >20 total sets (Baz-Valle synergist finding), which the indirect pressing supplies on top of the 14 direct.
Forearms (flexors/extensors/brachio radialis)	8	Covered by neutral-grip pulldown/row support-grip + EZ/hammer-pattern curls.

Muscle	Weekly sets	Note
Quadriceps	15 (Lower A ~10 — squat 4 + Bulgarian 3 + reclined leg ext 3; Lower B ~3 split squat 2°)	2×/wk. Knee-dominant heavy share on Lower A; deep-ROM squat for long-length loading; reclined/extended-hip leg extension biases the rectus femoris the 90°-hip squat misses.
Hamstrings	14 (Lower A ~3 long-length RDL; Lower B ~6 RDL + seated curl, split)	2×/wk via BOTH patterns: hip-extensor heads at long length (RDL / hip hinge) and knee-flexor heads (seated/lying curl). Seated curl biases the lengthened hip-extensor heads per Maeo.
Glutes (max + medius/minimus)	14 (Lower A ~7 squat/Bulgarian/RDL; Lower B ~8 RDL/thrust/split/abduction)	2×/wk, complementary length coverage: long-length via squat + RDL, short-length via Smith hip thrust (in for the short-length stimulus it ADDS, not as a squat replacement). Cable abduction for medius/minimus.
Adductors	8 (Lower A ~3, Lower B ~3 + deep-squat indirect)	Direct cable standing adduction through a deep abducted-stretch ROM, on both lower days; most stimulus also banked indirectly from deep squats.
Calves (gastrocnemius + soleus)	12 (Lower A ~4 standing/gastroc, Lower B ~4 seated/soleus)	Trained by knee angle (supported): standing / knee-extended for the biarticular gastroc (Lower A), seated / knee-flexed for the soleus (Lower B).
Core (rectus abdominis, obliques, erector spinae/QL)	10 (Lower A ~3, Lower B ~3 + braced-hinge isometric)	Anti-extension ab-wheel rollout to complement the braced squat/hinge; spans anti-extension + anti-lateral-flexion.

The 4 sessions

Upper A — Horizontal-Emphasis (Push-bias) + Shoulder-Health Block #1

Neutral/front-of-body horizontal pressing and rowing; lateral + rear delts; direct biceps/triceps; and the **FIRST** of two weekly programmed rotator-cuff + scapular-stability blocks. One heavier 6–8 rep neutral-grip press anchors the day; everything else lives 8–20. Posterior-girdle volume (row + rear delt + cuff) exceeds pressing volume per the structural-balance rule. Cuff/scap activation runs first to prime scapular stability and warm the joint, then the big compound press+row, then delts/arms/scap finisher.

#	Exercise	Sets × Reps	RIR	Load	Structure	Notes / accommodation
1	Cable External Rotation (elbow at side)	2 × 12–15	2–3	Cable, lightest pin (~10–15 lb)	straight-sets	Cuff/scap block 1 of 2. Elbow pinned at neutral abduction — well clear of the apprehension position; rotate out only to a comfortable, <i>stable</i> end-range, never forced. Slow controlled arc.
2	Supine Cable Scapular Protraction (serratus punch)	2 × 12–20	1–2	Cable, light (~15–20 lb)	straight-sets	Pure open-chain protraction with minimal glenohumeral motion — loads the serratus for scapular upward-rotation control with no joint stress. Programmed scap-stability work (RULE 6).

#	Exercise	Sets × Reps	RIR	Load	Structure	Notes / accommodation
3	Dumbbell Flat Bench Press (neutral grip) — <i>strength anchor</i>	4 × 6–8	3 early-block → 1–2 late	DB 60s start → toward the 75 lb cap; once 75×8 on all sets, progress via reps/tempo/ad ded set	straight-sets (cluster optional on the top set)	Day's heavier neutral-grip press. Neutral/semi-neutral grip, front-of-body, ~2 s controlled eccentric, NOT ballistic. Stop the descent where the shoulder feels stable — do NOT force the deep wide-grip end-range stretch. Keep 2–4 RIR, never grinding failure (intensity techniques banned here). <i>[neutral/front pressing — RULE 3; controlled tempo — RULE 6; 5–8 reps = genuine strength zone]</i>
4	Incline-Bench Chest-Supported DB Row (neutral grip)	4 × 10–12	2–3, last ~1	DB 45–55s	antagonist-superset with #3 (back-to-back, ~90–120 s shared rest)	Chest-supported = zero lower-back/cheat demand; neutral-grip, elbows drive back to full scapular retraction — directly balances the pressing volume. No end-range external rotation. <i>[antagonist superset, Burke 2025 ~36% time saved at equal gains; posterior ≥ pressing — RULE 7]</i>
5	Seated Cable Row (close neutral V-handle)	3 × 12–15	1–2, last ~1	Cable ~mid stack	straight-sets	Neutral V-handle, front-of-body, elbows tucked-and-back — wrist- and shoulder-friendly; pull to the lower ribs, let the scapulae protract at the stretch for full pain-free ROM. Second horizontal angle toward the 18 sets/wk back target. <i>[neutral-grip rows substituted for wide pronated end-range pulls — RULE 3]</i>
6	Cable Lateral Raise	4 × 12–20	1–2, final 0–1	Cable, light (~15–25 lb/side)	straight-sets (final set may extend to a myo-rep — cable isolation only)	Raise only to ~shoulder height in the frontal plane — no higher, keeping the humerus out of the abduction+ER apprehension zone. Constant-tension loaded stretch at the bottom. High-priority therapeutic lateral-delt volume (18/wk). <i>[intensity techniques only on cable/machine isolation — RULE 5]</i>
7	Cable Face Pull	4 × 15–20	1–2	Cable, light-mod (~20–30 lb)	straight-sets	Double-duty: rear delt + posterior cuff + mid-trap. Pull rope to forehead/nose height, elbows high but NOT driven behind the torso plane; external-rotate only to a comfortable, stable end-range. Central to both the rear-delt (16/wk) and cuff/scap (12/wk) priorities (RULE 6 + RULE 7).
8	Cable Cross-Body Rear Lateral Raise	3 × 15–20	1–2, final 0–1	Cable, light (~10–20 lb)	straight-sets	Loaded-stretch rear delt with the cuff external rotators assisting — horizontal abduction stays in front-to-side range, no behind-the-body end-range loading. Tops up rear delt toward the 16/wk balance safeguard (RULE 7).
9	EZ-Bar Curl	3 × 8–12	1–2, last 0–1	EZ-bar (≤60 lb), ~40–50 lb	antagonist-superset with #10 (~90 s shared rest)	Elbows in front of / beside the torso, no shoulder involvement — fully shoulder-neutral. EZ angle is wrist-friendly. Direct biceps toward the 14/wk target.
10	Rope Triceps Pushdown	3 × 12–15	1–2, last 0–1	Cable ~mid stack	antagonist-superset with #9 (final set may extend to a drop/myo-rep — cable isolation only)	Upper arms pinned to the sides, elbow-extension only — no shoulder loading. Direct triceps to complement the indirect pressing volume already banked from #3, toward 14/wk direct (>20 total favors triceps).

Shoulder rules encoded this day: neutral-grip/front-of-body presses only; NO behind-neck, NO deep dips, NO end-range stretch forced past a stable joint; ~2 s controlled eccentrics, no ballistic reps; intensity techniques kept OFF the press and confined to machine/cable isolation (#6, #10 only); the cuff + scap block (#1, #7, plus the serratus #2) runs at 1–2 RIR, controlled, never to grinding failure, treated as a programmed priority, not a finisher. **STOP-AND-REASSESS**

kill switch: halt on any subluxation, sharp pain, slipping/instability, arm numbness/tingling, or sudden strength loss — working-muscle pump/fatigue is fine.

Rest: neutral-press anchor 3 min; superset pairs ~90–120 s shared; cable isolation/cuff 45–90 s.

Lower A — Knee-Dominant (Quad-Emphasis)

Squat-pattern knee-dominant strength anchor (heavier 5–8), then quad-biased + rectus-femoris-biased + lengthened-hamstring + glute + adductor + **standing/gastroc** calf accessories at 8–25. Gastroc trained knee-EXTENDED here; soleus (seated, knee-flexed) is paired on Lower B. Shoulder stakes are low — the only cue is that the bar racks on the traps (front-of-neck) without forcing the arms into end-range external rotation at the unrack.

#	Exercise	Sets × Reps	RIR	Load	Structure	Notes / accommodation
1	High-Bar Back Squat — <i>strength anchor</i>	4 × 5–8	3–4 early → 2–3 late	Olympic bar ~225 lb working; cap ~570. Warm up to a 5–8RM leaving 2–3 RIR. +10 lb at the top of the band, reset to 5.	straight-sets (double progression)	Day's strength anchor. High-bar = bar on the upper traps, front-of-neck path, deep but pain-free ROM (no behind-neck). Self-spot the unrack so the hands rest comfortably without driving the elbows behind the torso into end-range ER. Controlled ~2 s eccentric, brace anti-extension. Squat to/below parallel for long-length quad/glute loading. <i>[deep ROM → more leg growth; straight sets = benchmark, lowest fatigue, best load/technique]</i>
2	Bulgarian Split Squat (dumbbell)	3 × 8–12 / leg	2, last ~1	DB 50s/hand (cap 75; then reps/tempo/add ed set)	straight-sets	Fairly upright torso to keep front-leg quad bias; deep front-knee ROM, back foot on a bench. DBs at the sides — no overhead/ER demand on the shoulder. <i>[added quad volume in the 12–20 MAV zone]</i>
3	Leg Extension (reclined / extended-hip, rectus-femoris bias)	3 × 12–20	1–2, final 0–1	Machine ~mid stack; last set as a drop off the working weight	drop-set on the last set only (one capped machine-isolation finisher); straight sets prior	Recline the seat / extend the hip to bias the biarticular rectus femoris (under-stimulated by the 90°-hip squat and standard leg extension). No shoulder involvement. <i>[train RF in a more hip-extended position; drop-set = machine-isolation time-efficiency only, not extra growth]</i>
4	Romanian Deadlift (RDL)	3 × 8–10	2–3	Olympic bar ~155–185 lb; hinge to a deep stretch (mid-shin), neutral spine	straight-sets (double progression)	Lengthened hip-extensor hamstring + glute stimulus on this knee-dominant day (knee-flexor curls concentrate on Lower B). Bar in front at arms' length — zero overhead/apprehension demand; brace the erectors, controlled eccentric. <i>[long-muscle-length stimulus rationale]</i>

#	Exercise	Sets × Reps	RIR	Load	Structure	Notes / accommodation
5	Cable Standing Hip Adduction	3 × 15–25 / leg	0–1	Cable ~low-mid via ankle-cuff; start each rep in a deep abducted stretch	straight-sets	This day's adductor share (weekly 8, split across both lower days). Ankle-cuff bypasses grip; no shoulder loading. Drive through a full lengthened-to-shortened arc.
6	Smith Machine Standing Calf Raise	4 × 8–15 (one back-off to 15–20)	1–2, final 0–1	Smith bar + ~90–135 lb on a block for full dorsiflexion stretch; cap ~550	straight-sets	STANDING / knee-EXTENDED to load the biarticular gastrocnemius at long length (soleus is trained seated on Lower B). Bar on the traps, front-of-body — no end-range shoulder position. Pause in the stretched bottom, full ROM. <i>[standing/knee-extended grows the gastroc more]</i>
7	Ab-Wheel Rollout	3 × 8–15	1–3	Bodyweight; regress to a shorter ROM if the anti-extension brace breaks	straight-sets	Anti-extension core to complement the braced squat/hinge (toward the weekly 10). Rolls in front of the body within a comfortable range — keep the shoulders stable, no end-range overhead reach.

Shoulder note (lower day): risk is minimal, but every loaded movement keeps the bar/implement **front-of-body with hands at the sides or on the traps** — no overhead, no behind-neck, no end-range external rotation / apprehension position. The therapeutic rotator-cuff + scapular block lives on the two UPPER days, not here. The red-flag kill switch still applies.

Gym note (290 Revolution Drive): there is **NO leg-press or hack-squat machine** in the menu, so the knee-dominant compound volume is built from the free-weight/Smith squat plus the dumbbell Bulgarian split squat rather than a machine press.

Rest: squat 3–4 min (up to 5 on the top set); Bulgarian/RDL 2–3 min; leg ext / adduction / calf / core 60–120 s. Optionally superset the cable hip adduction with the calf raise (non-competing) to trim session time.

Upper B — Vertical-Emphasis (front-of-body) + Shoulder-Health Block #2

Neutral/front-of-body overhead pressing and FRONT vertical pulling, with an incline neutral press for upper-pec angle variety, a repeat lateral + rear-delt dose, lengthened-bias arms, and the **SECOND** weekly rotator-cuff + scapular block. All pressing is neutral-grip, front-of-body, within pain-free ROM — no behind-neck, no deep dips, no ballistic overhead. Posterior-girdle + cuff/scap volume kept ≥ pressing (RULE 7) and dosed generously because upper training is therapeutic (RULE 8).

#	Exercise	Sets × Reps	RIR	Load	Structure	Notes / accommodation
1	Seated DB Shoulder Press (neutral grip) — <i>strength anchor</i>	4 × 5–8	2–4 (heavie st top set; never to failure)	DB 45s start → toward the 75 lb cap; once capped, reps/tempo/sets	straight-sets (double progression); cluster optional on the top set for velocity-maintained heavy pressing	Day's overhead anchor. Neutral (palms-facing) grip, press in front of the body within pain-free ROM; controlled ~2 s eccentric, no jerk/push-press; stop just shy of lockout end-range, never let the elbows drift behind the torso plane. <i>[failure not required — heavy press lives at 2–4 RIR; RULE 3/4/6]</i>
2	Close Neutral-Grip Lat Pulldown (to chest)	4 × 10–15	1–2, last 0–1	Machine ~mid stack, to the upper chest	antagonist-superset with #1 (~90–120 s shared rest)	Neutral-grip handle, bar to the FRONT/upper chest only — never behind the neck; full pain-free overhead stretch without forcing end-range. <i>[antagonist superset, RCT ~36% less session time; RULE 3/4]</i>
3	Incline DB Bench Press (neutral grip)	3 × 8–12	2–3, last ~1	DB 40–50s on a ~30° incline	straight-sets (double progression)	Neutral/semi-neutral grip, front-of-body; keep the stretch full but pain-free — do not force the deep wide-grip bottom stretch at the shoulder. Adds the upper/clavicular pec angle. <i>[incline upper-pec angle — modest edge; RULE 3/5]</i>
4	Cable Lateral Raise	3 × 12–20	0–2, final 0–1	Cable ~low-light, constant tension w/ loaded stretch	myo-reps on the last set (machine/cable isolation only); antagonist-superset with #5	Raise in the frontal/scapular plane to ~shoulder height only — no end-range elevation that drives ER in abduction (stay out of the apprehension zone). High therapeutic priority, dose generously (RULE 8). <i>[myo-reps — RCT comparable growth, less time, better recovery]</i>
5	Cable Face Pull	3 × 12–25	1–2	Cable ~light-mid, rope to forehead height, elbows high	antagonist-superset with #4 (~90–120 s shared rest)	External rotation only WITHIN available ROM — pull to the face, stop where the cuff feels stable; controlled tempo, no grinding (programmed cuff/scap block, RULE 6). Keeps posterior volume ≥ pressing (RULE 7).
6	Cable External Rotation (elbow at side)	3 × 12–20	1–3 (never grindin g failure)	Cable ~lightest practical, elbow pinned to side	straight-sets, 45–75 s rest, low-fatigue	Dedicated SECOND weekly rotator-cuff dose. Elbow-at-side angle (NOT 90° abduction) keeps the cuff out of the apprehension position; rotate out only to the pain-free end of available ER ROM, controlled tempo. <i>[lowest-fatigue benchmark for cuff/shoulder-stakes work; RULE 6]</i>
7	Overhead Shrug (scapular upward-rotation shrug)	2 × 10–20	1–3	DB 20–30s overhead (or light Smith bar), short controlled range	straight-sets	Trains the serratus + trap upward-rotation force couple that stabilizes the scapula for overhead pressing; keep load light and the path controlled within pain-free overhead ROM — no end-range strain (RULE 6).
8	Incline DB Curl + Overhead Cable Rope Triceps Extension	3 each × 10–20	1–2, final of each 0–1	Curl: DB 20–30s on incline (long-head stretch). Triceps: cable ~light-mid, rope overhead (long-head stretch)	antagonist-superset (biceps + triceps back-to-back, ~90 s shared rest); myo-reps optional on the curl's last set	Both loaded in the lengthened (stretched) position; the overhead triceps stretch stays within a comfortable, pain-free shoulder-flexion range — no forced end-range overhead (RULE 5). <i>[lengthened curls modestly out-grow shortened; antagonist superset]</i>

Shoulder rules encoded this day: neutral-grip/front-of-body overhead + incline pressing only; FRONT-to-chest pulldown, never behind-neck (RULE 3/4); the SECOND programmed cuff/scap block (#5 face pull + #6 cable ER + #7 overhead/serratus shrug) at 1–2 RIR, controlled, treated as a priority; intensity techniques (myo-reps) confined to #4 cable lateral raise and the #8 curl — never the press (RULE 5). ROM rule on every movement: full but PAIN-FREE — hard stop at the first sign of instability/apprehension; STOP-AND-REASSESS on any red flag.

Rest: ~3 min on the press; ~90–120 s shared on supersets; 45–75 s on cuff/scap work.

Lower B — Hip-Dominant (Hamstring / Glute-Emphasis)

Hip-hinge strength anchor (heavy 5–8 RDL) into a glute-max short-length thrust complex, then knee-flexor/lengthened-hamstring curl, quad-biased split squat, adductor + glute-medius accessories, a **soleus-biased** second calf session (the head not emphasized Lower A), and braced anti-extension core. Shoulder-neutral throughout: every load is front-of-body / neutral-grip / grip-free with no overhead or end-range positions, so this lower day adds zero apprehension-position exposure.

#	Exercise	Sets × Reps	RIR	Load	Structure	Notes / accommodation
1	Romanian Deadlift (RDL) — <i>strength anchor</i>	3 × 5–8	3 early → 2 late (top set to 1 end-of-block only)	Olympic bar ~155–185 lb start; cap ~570 (loadable indefinitely). Straps once grip is the limiter so hamstrings drive failure.	straight-sets (double progression); reverse-pyramid is an optional autoregulatory swap on this anchor	Front-of-body, neutral-grip hold — fully shoulder-safe. Brace the lats, keep the bar against the legs; hinge to a hard hamstring stretch, no rounding, controlled ~2 s eccentric. <i>[straight sets = benchmark, no technique out-grows it at matched volume; high-SFR hamstring/glute hinge at long muscle length]</i>
2	Smith Machine Hip Thrust	3 × 8–12	2–3, last ~1	Smith ~bar+90–135 lb start; cap ~550. Pad across the hips; full lockout, 1 s glute squeeze	straight-sets (double progression)	Grip-free, no shoulder loading — bar sits on the hips. Ribs down, chin tucked, posterior pelvic tilt at lockout; no low-back hyperextension. <i>[short-length glute stimulus that complements the RDL's long-length loading — the two are programmed together for complete glute coverage, not because the thrust out-grows the squat/hinge]</i>
3	Seated Leg Curl	3 × 12–20	1–2, final 0–1	Machine ~mid stack	myo-reps optional when time-limited; otherwise straight-sets. Antagonist-superset with #4 to compress time	Seated (hip-flexed) trains the hamstring at long length, biasing the hip-extensor heads and stretch-mediated growth; no shoulder involvement. <i>[myo-reps — RCT-comparable growth, better recovery on isolation; seated-curl long-length bias]</i>
4	Bulgarian Split Squat (dumbbell)	3 × 8–12 / leg	1–2	DB 30–40s/hand start (cap 75; then reps/tempo/sets). Slight forward lean to share load into glute + adductor	straight-sets (double progression); antagonist-superset with #3 (~36% time saved)	DBs at the sides (neutral, front-of-body) — no shoulder demand. Covers the day's secondary quad target plus extra glute-max long-length and adductor work.

#	Exercise	Sets × Reps	RIR	Load	Structure	Notes / accommodation
5	Cable Standing Hip Adduction	3 × 15–30 / leg	0–1	Cable + ankle-cuff ~low-mid; working leg starts in a deep abducted stretch	straight-sets	Grip-free cable accessory; loads the adductors through a long abducted-stretch ROM. No upper-body / shoulder demand.
6	Cable Standing Hip Abduction	2 × 15–30 / leg	0–1	Cable + ankle-cuff ~low; lead leg out-and-slightly- back to bias glute medius	straight-sets (minimal-fatigue, short rest)	Grip-free, very low systemic cost; covers the glute medius/minimus abduction target. No shoulder loading.
7	Smith Machine Seated Calf Raise	4 × 15–30	0–1	Smith + plates ~90–135 lb across the knees; pause 1 s at full stretch, full plantarflexion at top	straight-sets; optional capped machine drop-set on the last set	SEATED, knee-bent — biases the SOLEUS , the calf head NOT emphasized in Lower A (which trained standing/gastroc). No upper-body involvement. <i>[seated knee-flexed raises train the soleus; Lower A's standing raise covers the gastroc]</i>
8	Ab-Wheel Rollout	3 × 8–15	1–2	Bodyweight; regress to a kneeling partial range if the lengthened position is too hard, extend as it builds	straight-sets	Anti-extension core to match the braced-hinge demand; arms stay in front of the body, no overhead or end-range shoulder position. Keep ribs down, avoid lumbar sag.

Shoulder note (lower day): every implement — RDL bar, Smith hip-thrust/calf bar, side-held dumbbells, ankle-cuff cables, ab wheel — is **front-of-body or grip-free** with no overhead, behind-neck, deep-dip, or abduction-plus-external-rotation exposure. The apprehension position is fully avoided. The red-flag kill switch still applies: stop and reassess on any subluxation, sharp joint pain, slipping sensation, arm numbness/tingling, or sudden strength loss.

Rest: RDL 3–4 min (heavy hinge anchor); hip thrust / split squat 2–3 min; leg curl / adduction / abduction / calves 60–120 s; ab rollout ~90 s. Antagonist-superset option: pair the seated leg curl (#3) with the Bulgarian split squat (#4) back-to-back for ~36% time savings without losing volume.

Dynamic mesocycle (the progression)

Length: one 4–5 week accumulation block + a deload, run as a continuous mesocycle on top of the static Upper/Lower split. Volume ramps from MEV into the MAV zone while RIR descends ~4 → ~1 (Martikainen — equal gains at lower average session-RPE); the **rotator-cuff + scapular block runs at full 12 sets/week from week 1 and never ramps** (it is therapeutic baseline, not a progression variable). Run the optional 5th week only if recovery is still tracking; collapse to a 4-week block + deload if performance stalls in week 4.

Week	Focus	Volume	Intensity (RIR)
Wk 1 — Accumulation entry / re-groove	Re-groove all four strength anchors (Upper A neutral flat press, Lower A squat, Upper B neutral overhead/incline, Lower B RDL/hip-thrust) with full technique focus; on the shoulder, confirm the neutral/front-of-body press groove is pain-free through full available ROM. Both cuff/scap blocks run at full dose from day 1. Straight-sets only this week (clean reference for the double-progression); introduce antagonist supersets on the non-competing upper pairs once movements feel grooved.	START at MEV, biased generous on the therapeutic targets. Roughly: lateral delt ~14, rear delt ~12, back ~14, chest ~11, biceps/triceps ~11 each, quads ~12, hams ~11, glutes ~11, calves ~10, plus the fixed 12-set cuff/scap block at full dose (RULE 8 — therapeutic, not protective under-dosing). Lower body and arms enter at the bottom of their range; shoulder-health and lateral/rear-delt work enter higher because their MEV is effectively their working dose.	Top of the descending ramp. Compound anchors ~4 RIR (RPE 6); primary hypertrophy compounds 3 RIR; isolation 2 RIR. Heavy anchors sit high deliberately — failure on big lifts is slightly inferior for strength and ~triples acute neuromuscular fatigue vs 3 RIR.
Wk 2 — Accumulation, first double-progression step	On every exercise, attempt the top of its rep band at week-1 load before adding the smallest increment (DBs cap at 75 lb — then reps/added sets/tempo, never grip-width creep on the shoulder). Strength anchors begin loading at the heavy 5–8 end. Antagonist supersets now standard: horizontal-row + neutral-press, rear-delt/face-pull + lateral-raise, biceps + triceps, seated-leg-curl + split-squat.	Ramp + 1–3 sets into PRIORITY muscles only (lateral delt, rear delt, back, triceps) — not lower body. Target ~ lateral delt 16, rear delt 14, back 16, triceps 12; hold lower body and chest. Cuff/scap stays pinned at 12. Stay inside the 12–20 trained optimum (diminishing returns past ~20).	Anchors ~3–4 RIR; primary hypertrophy compounds 2–3 RIR; isolation 1–2 RIR with the last set of each isolation to 0–1 ; any light-load high-rep isolation set to 0–1 (recruitment is recruitment-late at low loads).
Wk 3 — Accumulation peak / push toward MAV	Continue double progression — most exercises now adding load or sitting mid-band after last week's add. Bring the remaining muscles up to target. Introduce time-efficient intensity structures ONLY on machine/cable isolation : myo-reps or one capped drop-set on lateral raise / leg extension / calf / triceps pushdown — kept OFF every shoulder-loaded and free-weight lift (RULE 5). Optional cluster sets on the heavy neutral-grip press for a top-end strength bias (velocity-maintained, lower fatigue, keeps technique crisp on the unstable shoulder).	Near top of the MAV climb : lateral delt ~18, rear delt ~16, back ~18 (the generous therapeutic ceilings), chest ~13–14, quads ~14–15, hams ~13–14, glutes ~13–14, arms ~13–14, calves ~12. Brushes the upper end of the 12–20 optimum — do NOT exceed absent a clear MRV headroom signal. Cuff/scap remains pinned at 12.	Anchors ~3 RIR; primary hypertrophy compounds ~2 RIR, last set ~1; isolation 1 RIR, final sets to 0–1 (0 on light-load high-rep finishers). RIR continuing its descent.

Week	Focus	Volume	Intensity (RIR)
Wk 4 — Intensification (effort over added volume)	Volume HOLDS at the week-3 ceiling; the progression lever shifts to LOAD on the strength anchors and proximity-to-failure on hypertrophy work. The genuine 5–8 strength zone is expressed: anchors push load at the heavy end, optionally via reverse-pyramid on the LOWER-body anchors only (heaviest top set first while fresh; AVOIDED on overhead/unstable-shoulder pressing where a warmed-up straight-set ramp is safer). Watch for MRV signals and hold any muscle that flags one.	HOLD volume at week-3 levels; redirect adaptation pressure into intensity, not more sets. If a muscle missed performance vs week 3, drop IT back ~2–3 sets while leaving the others — never a blanket cut. If fatigue forces a trim, the cuff/scap block and lateral/rear-delt therapeutic volume are the LAST things touched (RULE 8).	Anchors ~2–3 RIR (reserve 0–1 for an end-of-block top set, and only on lower-body / neutral-press lifts that don't compromise the joint); primary hypertrophy compounds ~1–2 RIR, last set failure-adjacent; isolation 0–1 throughout.
Wk 5 — Intensification peak (optional)	Run it only if recovery is still tracking. Final realization: top-end loads on the four anchors at their lowest planned RIR, hypertrophy work at peak proximity. No new structures — an expression week, so shoulder-stakes lifts stay on plain straight-sets/clusters at controlled non-ballistic tempo (RULE 6). End-of-block top sets are the ONLY place 0–1 RIR is permitted on a heavy press, and even then only the neutral/front patterns Aaron presses pain-free.	Still HELD at the block ceiling (lateral delt ~18, rear delt ~16, back ~18, etc.) — a brief touch of MRV to trigger supercompensation, not a new volume PR. Expect most muscles to finally signal MRV (can't beat prior performance) — that's the trigger to deload, not to add sets.	Bottom of the ramp (~RIR 1 average). Anchors ~2 RIR (a single end-of-block top set may touch 0–1 on safe patterns only); primary hypertrophy compounds ~1; isolation 0–1 / final sets to failure. 1–3 RIR already captures nearly the full hypertrophy stimulus — do not grind every set to 0.
Deload (wk 6 after a 5-wk block; wk 5 after a 4-wk block)	A fatigue flush, not a layoff — keep the movement patterns and the joint moving (avoiding the sedentary state that flares the shoulder); just drop the dose.	TWO LEVERS, RIR FIRST. (1) Raise effort to 3–6 RIR on everything (the higher-value, lower-cost lever — removes the fatigue-generating last reps). (2) Cut volume toward MV (~6 sets/muscle) on the high-fatigue work: heavy lower-body anchors (squat/RDL/hip-thrust) and the heaviest pressing, halved or dropped to ~6 sets, loads pulled back ~10–20%, reps submaximal. CRITICAL ASYMMETRY — protect the therapeutic upper dose: do NOT cut the upper body to MV uniformly. The cuff/scap block stays at/near its full 12 sets (low-fatigue, never to grinding failure, the literal mechanism that resolves Aaron's symptoms). Lateral/rear-delt and back work drop only modestly (~8–10 sets), not to 6. After the deload, restart at MEV with RIR ~4 and re-run the ramp.	3–6 RIR, no failure anywhere. The deload is a fatigue-management tool, not an independent growth driver — its job is to let the next block start fresh.

Autoregulation & how to read this

Day-to-day / week-to-week adjustments — ordered so upper-body therapeutic volume is the LAST thing cut, never the first.

- **RED-FLAG KILL SWITCH (overrides everything, RULE: stop-and-reassess).** STOP the set and reassess immediately on any of — a dislocation/subluxation event, sharp pain, a sense of the shoulder slipping/instability, numbness or tingling

down the arm, or sudden loss of strength. These are joint-line/neural signals, categorically different from the dull working-muscle fatigue/pump that is expected and fine. A red flag ends *that exercise* for the day; it does NOT mean cut the week's upper-body volume.

- **WHEN THE SHOULDER FEELS TIGHT (but no red flag).** This is a cue to MODIFY THE MOVEMENT, NOT remove volume — in-ROM upper training is what resolves Aaron's symptoms (RULE 8). In priority order: (1) shrink ROM to the pain-free/stable portion and keep the set; (2) swap the loaded pattern for a friendlier one in the same muscle target — neutral-grip / front-of-body / machine or cable instead of free-weight, landmine instead of straight-bar overhead (RULE 1); (3) slow the tempo and drop the load, keeping 2–3 RIR (RULE 4, no ballistic reps); (4) front-load that day's cuff + scap block as a "feeder" to warm and stabilize the joint, then re-attempt the pressing. Only AFTER these fail do you reduce that single exercise's sets — and you bank the removed sets into the cuff/scap or rear-delt work, keeping the girdle balanced (RULE 7).
- **WHEN RECOVERY IS POOR (bad sleep, high life stress, can't match last session's reps/load at adequate rest = an MRV signal).** CUT EFFORT BEFORE VOLUME, AND CUT THE RIGHT VOLUME FIRST. (1) First lever — raise RIR by ~1–2 across the session (anchors to 3–4, isolation to 2); this recovers cheaply and 1–3 RIR already captures nearly all the hypertrophy stimulus. (2) Second lever — if still flat, trim sets from the HIGHEST-FATIGUE work FIRST: the heavy lower-body anchors and the heaviest pressing, and drop the optional intensity techniques (drop-sets/myo-reps) before anything else. (3) PROTECTED LAST — the cuff/scap block, lateral/rear-delt, and back volume are the final things touched; the cuff/scap block is effectively never cut on a bad week (low-fatigue, therapeutic). A bad week is exactly when the sedentary-flare risk is highest, so the dose that keeps the shoulder healthy must survive.
- **WHEN PERFORMANCE STALLS ACROSS A WHOLE MUSCLE for 2+ sessions despite rest:** that muscle has hit MRV — hold or shave 2–3 sets from THAT muscle only (not a blanket cut), or trigger the deload if multiple muscles flag together.
- **DEFAULT STRUCTURE BIAS UNDER FATIGUE:** when time or recovery is short, keep the shoulder-stakes compounds on plain straight-sets (lowest fatigue, best load/technique control, and the structure no advanced technique out-grows at matched volume) and shed the time-saving structures rather than the safe ones. Use antagonist supersets to preserve total volume in less time on a rushed day BEFORE cutting any sets.

How to read the tables. - **Sets × Reps** is the working prescription after warm-ups (warm-up ramps are not counted as working volume). - **RIR** = Reps In Reserve at the *target* point in the block; the early-block / late-block split shows the descending ramp (e.g. "3 early → 1–2 late"). - **Load** is a conservative week-1 starting point and the progression rule for that lift. **DBs cap at 75 lb** — once a DB lift tops its band at 75 on all sets, progress via reps / tempo / added set / structure, not load; barbell and Smith are effectively uncapped at this gym. - **Structure** names the set-structure used and (where relevant) which exercise it supersets with. *Straight-sets is the default and anchors every shoulder-stakes compound*; intensity techniques (drop / myo-reps / cluster) appear only where the Notes permit, and never on a shoulder-loaded press. - **Notes / accommodation** carries the cue, the shoulder rule encoded, and the inline citation justifying the choice.

Constraint accommodations baked in (always on). - **Bilateral shoulder instability (LIMITED external-rotation ROM, no pain):** neutral-grip / front-of-body pressing only; apprehension position (abduction + end-range ER) banned → no behind-neck press or pulldown, no wide end-range overhead, no deep dips, no forced end-range stretch under load; controlled non-ballistic tempo; intensity techniques kept off all shoulder-loaded lifts; a programmed rotator-cuff + scapular block twice weekly at full dose; posterior-girdle volume ≥ pressing volume. **The protection IS the training** — upper volume is dosed generous-but-recoverable, because sedentariness worsens his symptoms and in-ROM training resolves them. - **Red-flag kill switch:** stop and reassess on any subluxation/dislocation, sharp pain, slipping/instability sensation, arm numbness/tingling, or sudden strength loss. Working-muscle pump/fatigue is fine; joint-line/neural signals are not. - **Gym (290 Revolution Drive):** no leg-press / hack-squat → knee-dominant volume from free-weight/Smith squat + DB Bulgarian split squat; DBs cap at 75 lb → progress maxed DB lifts via reps/tempo/structure or migrate to barbell/Smith/machine.

Reminder: a separate document covers Aaron's Planet Fitness routine. This file is the 290 Revolution Drive (Somerville) routine only.